0/6 Questions Answered

Project initial implementation - DevOps/platform team member

Q1 Joint responsibilities

1 Point

Please upload:

A pair of JSON files representing

- one input sample (with real representative values)
- and one model output (again, with real representative values).

(Training, serving, and data team members must agree on this item together; training and inference team members ingest an input like the sample and produce an output like the sample, data team member produces something like the input sample in online and offline workflows.)

If your project involves more than one model, you will have one JSON pair per model. In that case, please name each JSON clearly to indicate which model it "belongs" to.

Input JSON sample(s):

Please select file(s) Select file(s)

Output JSON samples(s):

Please select file(s) Select file(s)

Also upload

- a table enumerating all the containers involved in each role, with links to their Dockerfiles/Docker Compose files, and a link to the equivalent K8S manifest for each one.

The object is to show that each of the individual role-owned systems will be supported by the DevOps/Platform role, although at this stage team members are working independently.

> **Upload in PDF format.**

Please select file(s) Select file(s)

Save Answer

Q2 DevOps/Platform contributions

9 Points

Requirements to get credit for those deliverables:

- Kubernetes is required for 4-person teams. (For 3-person teams, Kubernetes is optional; Docker Compose is acceptable.)
- Git as source of truth: IaC/CaC artifacts and Kubernetes manifests (or equivalents) are in the repo.
- Durability: platform state and artifacts persist across pod restarts (MLflow artifacts and other shared artifacts use a persistent volume/object storage, not ephemeral container filesystems).
- Secrets hygiene: no secrets in Git.

Q2.1 Infrastructure requirements table:

1 Point

For each service running in your cluster, show the GPU, CPU, memory requests and limits you set, plus brief evidence from Chameleon showing how you arrived at appropriate values for right-sizing.

Upload PDF document.

Please select file(s) Select file(s)

Save Answer

Q2.2 Repository artifacts

4 Points

Please upload:

- IaC/CaC materials that provision Chameleon infrastructure and configure a Kubernetes cluster (cluster, networking/ingress, persistent volumes, namespaces).
- K8S manifests and other necessary materials to deploy the open source service on which the project is based, and to deploy platform services required by other team members.

Create a `.zip` or `.tgz` file out of your "infrastructure" repository (or repositories, if you have more than one), and upload it here:

Please select file(s) Select file(s)

Use this text field to explain how to use it, i.e. what is the sequence of steps to bringing up your system?

Note: There is no credit for systems that are not running on Chameleon Cloud. Code or configuration that has not been executed in the target environment, or that only runs locally, does not count. You should *only* submit items that you have implemented and executed on Chameleon.

Save Answer

Q2.3 Demo video of open source service
2 Points

Upload a sped-up demo video of the selected open source service running inside Kubernetes on Chameleon.

(Video should demonstrate everything from launching the service to confirming its health status inside K8S to validating in a browser that it is reachable and functional.)

The video should be no longer than a few minutes.

Please select file(s) Select file(s)

Note: There is no credit for systems that are not running on Chameleon Cloud. Code or configuration that has not been executed in the target environment, or that only runs locally, does not count. You should *only* submit items that you have implemented and executed on Chameleon.

Save Answer

Q2.4 Demo video of shared platform services
2 Points

Upload a sped-up demo video of shared platform services running inside Kubernetes on Chameleon, with persistent storage as appropriate.

(Video should demonstrate everything from launching the service to confirming health status inside K8S to validating in a browser that it is reachable and functional.)

The video should be no longer than a few minutes.

Please select file(s) Select file(s)

Note: There is no credit for systems that are not running on Chameleon Cloud. Code or configuration that has not been executed in the target environment, or that only runs locally, does not count. You should *only* submit items that you have implemented and executed on Chameleon.

Save Answer

Q3 Bonus
0 Points

If you are attempting the extra credit:

Integrate a platform tool/framework not used in the lab assignments that materially improves operability, and justify why it improves your design using a realistic example. You may investigate frameworks for secrets management, TLS automation, centralized logging, image security/scanning, distributed tracing, etc. (Note: Prometheus/Grafana do not "count" because they are used in Lab 8.) You must show one concrete operational win in demo video + a short justification.

Submit the demo video and PDF document explaining, including the concrete realistic example, and also any artifacts (e.g. Dockerfiles, Python scripts) that go along with it.

Please select file(s) Select file(s)

Note: There is no credit for systems that are not running on Chameleon Cloud. Code or configuration that has not been executed in the target environment, or that only runs locally, does not count. You should *only* submit items that you have implemented and executed on Chameleon.

Save Answer

Save All Answers

Submit & View Submission